Mathematical Symbols

Symbol	**How to read it**	**Notes on meaning and usage**
$a = b$	a equals b	a and b have exactly the same value
$A \equiv B$	Equal by definition	\$ $1 \equiv 100$ pennies
$P \Rightarrow Q$	P implies Q	If P true, then Q true; Don't confuse with $\rightarrow$
$f : X \rightarrow Y$	Function f maps set X into set Y	Every function is a mapping
$P \Leftrightarrow Q$ P iff Q	P equivalent to Q, or P if and only if Q	P and Q imply each other
(a, b)	Open interval from a to b	Points between a and b, not including a and b
$[a, b]$	Closed interval from a to b	Points between a and b, including a and b
$\mathbb{R}$	The real numbers	*All* numbers either positive, negative or zero
$\mathbb{Z}$	The integers	$..., -2, -1, 0, 1, 2, 3, ...$
$\mathbb{N}$	The natural numbers	$1, 2, 3, 4, ...$
$\mathbb{R}^n$	n-dimensional vector space over $\mathbb{R}$	Ex: $(2, 4.1, 0.2, 1, 3) \in \mathbb{R}^5$
$a \in B$	a is an element of B	Not to confuse with $\subset$, which refers to set inclusion or ϵ, which is the Greek letter epsilon
$a \notin B$	a is not an element of B	
$A \subset B$	Set A is a strict subset of B	Anything in A is also in B and $A \neq B$
$A \subseteq B$	Set A is a weak subset of B	Anything in A is also in B, and possibly $A = B$
$A \cup B$	A union B	Set of all points that are either in A or B or both
$A \cap B$	A intersection B	Set of all points that are both in A and B
$\exists$	There exists	Just shorthand
s.t.	Such that	Shorthand. Sometimes also written as :
$\forall x$	For all x	Ex: $\forall x > 0$, $\exists y \in \mathbb{R}$ s.t. $y < x$ and $y > 0$
$\exists!$	There exists a unique	$\exists!\ x : x = 4.78943$
$\emptyset$	The empty set	The set containing nothing (scary...)
2^X	Set of all subsets of X, also called the power set of X	$X = \{a, b, c\} \Rightarrow$ $2^X = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$
$\{\cdots\}$	An unordered set of things	Ex: $\{a, b, c\} = \{c, a, b\} = \{b, c, a\}$, etc.
$(\cdots)$	A vector, or ordered set	Ex: (a, b, c) as a point in 3-D space
$X \setminus Y$	X minus all elements of Y	Ex: $2^{\{a,b,c\}} \setminus 2^{\{a,b\}} = \{\{c\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$
$\lvert X \rvert$	Cardinality of X (# elements in X)	$\lvert\{a, b, c, d\}\rvert = 4$